

ARLINGTON, WA, USA

WMO: 727945

Lat: 48.161N

Lon: 122.159W

Elev: 42

StdP: 100.82

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 04205

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.0	-4.6	-10.2	1.6	-4.6	-7.6	2.0	-3.0	11.8	9.9	10.7	9.6	0.5	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	28.5	19.3	27.0	18.3	24.8	17.2	19.7	27.3	18.7	25.8	17.8	24.2	4.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.7	22.2	15.9	11.4	21.9	14.7	10.5	21.0	55.9	27.3	52.7	26.0	49.8	24.0	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.1	7.0	DB	-9.6	31.7	3.1	2.2	-11.8	33.3	-13.6	34.6	-15.4	35.9	-17.6	37.5	(4)
(5)			WB	-10.2	21.5	3.2	1.2	-12.4	22.4	-14.3	23.1	-16.1	23.7	-18.4	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	4.7	4.8	7.0	9.3	12.2	14.7	17.3	17.5	14.9	10.2	6.4	3.6
	DBStd	5.65	3.79	3.30	2.93	2.58	2.81	2.50	2.40	2.18	2.52	2.88	3.55	3.60
	HDD10.0	814	166	147	100	42	10	1	0	0	1	33	117	199
	HDD18.3	2993	421	379	351	272	192	112	49	42	107	253	359	456
	CDD10.0	904	3	1	8	21	77	142	226	233	147	39	7	1
	CDD18.3	41	0	0	0	0	1	4	17	17	3	0	0	0
	CDH23.3	730	0	0	1	6	23	81	267	269	83	1	0	0
	CDH26.7	167	0	0	0	1	3	17	74	60	13	0	0	0
Wind	WSAvg	2.3	2.4	2.3	2.8	2.6	2.2	2.2	2.2	2.0	1.9	2.0	2.4	2.2
Precipitation	PrecAvg	1185	138	94	115	99	88	69	30	31	60	119	178	159
	PrecMax	1459	247	180	186	152	155	131	89	103	128	213	306	244
	PrecMin	940	72	14	46	33	31	17	1	0	17	36	69	90
	PrecStd	158	41	40	33	25	30	32	20	28	32	53	62	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	15.0	19.7	23.0	26.2	28.9	32.2	30.9	28.0	21.4	16.4	13.8
		MCWB	11.2	9.7	12.2	15.2	17.3	19.7	20.5	19.6	18.3	14.5	13.2	11.3
	2%	DB	12.6	12.7	16.2	19.0	22.7	26.1	28.7	28.0	26.1	18.8	14.0	11.4
		MCWB	10.1	8.8	10.7	13.1	15.8	18.0	19.5	18.9	17.4	13.9	11.5	9.4
	5%	DB	11.5	11.3	13.9	17.1	20.9	22.9	26.3	26.4	23.0	17.2	12.7	10.0
		MCWB	9.4	8.2	9.6	11.6	15.0	16.2	18.0	18.1	16.2	13.1	10.5	8.3
	10%	DB	10.1	9.8	12.4	14.5	17.9	21.1	23.9	24.0	21.3	15.2	12.0	8.7
		MCWB	8.2	7.6	9.0	10.0	13.3	15.4	17.2	17.2	15.5	11.9	10.0	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.0	13.0	15.9	17.9	20.3	21.3	20.6	19.5	16.4	14.0	11.7
		MCDB	13.5	13.3	18.4	22.0	24.3	27.9	29.8	28.3	26.5	19.4	16.2	13.2
	2%	WB	10.5	9.6	11.5	13.4	16.5	18.3	19.7	19.4	18.0	14.5	12.1	9.7
		MCDB	12.1	11.8	15.2	17.8	22.6	25.4	27.5	27.1	24.5	17.5	13.6	11.1
	5%	WB	9.3	8.5	10.3	11.9	15.2	16.7	18.6	18.6	16.9	13.3	10.9	8.2
		MCDB	10.9	10.6	13.4	15.9	20.0	22.5	25.6	25.8	22.8	16.4	12.7	9.4
	10%	WB	8.1	7.7	9.1	10.6	13.8	15.5	17.5	17.6	15.9	12.3	9.9	7.4
		MCDB	9.5	9.3	12.0	14.1	18.0	20.1	23.6	23.9	20.6	14.8	11.6	8.4
Mean Daily Temperature Range	5% DB	MDBR	6.5	8.2	8.9	9.6	10.1	10.3	12.2	12.6	11.7	9.1	7.5	6.4
		MCDBR	7.6	10.8	12.6	14.5	15.2	15.4	16.7	16.8	16.2	12.2	8.2	6.6
		MCWBR	5.6	7.0	8.0	9.1	9.2	8.9	8.8	8.7	9.3	8.1	6.0	5.1
	5% WB	MCDBR	6.6	8.2	10.9	12.3	13.7	14.3	15.5	15.4	15.0	10.4	6.8	5.8
		MCWBR	5.2	5.9	7.3	8.1	8.6	8.5	8.6	8.4	8.8	7.4	5.3	4.8
Clear-Sky Solar Irradiance	taub	0.311	0.307	0.320	0.347	0.368	0.355	0.346	0.350	0.336	0.325	0.311	0.313	
	taud	2.525	2.546	2.538	2.451	2.414	2.468	2.514	2.513	2.546	2.572	2.559	2.466	
	Ebn at Noon	772	865	905	905	894	907	910	893	876	830	767	712	
	Edh at Noon	61	73	86	103	112	106	100	96	85	71	59	58	
All-Sky Solar Radiation	RadAvg	0.98	1.86	2.84	4.26	5.37	5.90	6.60	5.59	4.00	2.24	1.15	0.79	
	RadStd	0.14	0.24	0.33	0.40	0.43	0.56	0.48	0.37	0.33	0.26	0.16	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page